

BEYOND THE PROMPT.

Programming as a Pathway to Statistical Thinking

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Nonpartisan, independent and nonprofit, our mission is to serve as the trusted authority on money in American politics. We pursue our mission by providing comprehensive and reliable data, analysis and tools for policymakers, storytellers and citizens. Our vision is for Americans to use data on money in politics to create a more vibrant, representative and responsive democracy.

We pursue our mission with three goals:

- **Data:** We collect, clean and create access to quality and timely data on money in politics for an informed public.
- **Insights:** We generate and publish regular analysis, reports and user-friendly tools for people to interpret our data.
- **Engagement:** We fuel a growing community of policymakers, storytellers, and citizens who follow the money in American politics.

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Foreign-Connected PACs • OpenSecrets.org

opensecrets.org/political...

Foreign Connected PACs, 2007-2008

Export to CSV

Filter results

PAC Name (Affiliate)	Country of Origin/Parent Company	Total	Dems	Repubs
7-Eleven	Japan/Seven & I Holdings	\$9,500	\$2,000	\$7,500
ABB Group	Switzerland/Asea Brown Boveri	\$18,000	\$3,000	\$10,000
Accenture (Accenture)	UK/Accenture plc	\$347,750	\$189,750	\$158,000
Accenture (Accenture)	UK/Accenture plc	\$0	\$0	\$0
ACE INA	Switzerland/ACE Ltd	\$171,500	\$82,500	\$89,000
Acuson Corp (Siemens AG)	Germany/Siemens AG	\$0	\$0	\$0
AEGON USA	Netherlands/Aegon NV	\$210,497	\$107,997	\$102,500
Agrium US	Canada/Agrium Inc	\$1,500	\$0	\$1,500
Air Liquide America	France/L'Air Liquide SA	\$11,000	\$3,000	\$3,000
Alcatel-Lucent				

DONATE

Open Secrets

Only American citizens (and immigrants with green cards) can contribute to federal politics, but the American divisions of foreign companies can form political action committees (PACs) and collect contributions from their American employees. (Source: <https://www.opensecrets.org/political-action-committees-pacs/foreign-connected-pacs/2024>).

In this exercise you will work with data from contributions to US political parties from foreign-connected PACs. The data is stored in CSV files in the `data` directory of your repository/project. There are 13 files, each for an election cycle between 2000 and 2024.

a. Load all the data at once and stack them on top of each other with a column indicating which year the data is from with a single call to `read_csv()`. Call the data frame `pac_raw`.

b. Clean the names of the variables in the dataset with a new function from the janitor package: `clean_names()`. Then clean and transform the data such that you have contribution amounts to Democrats and Republicans as separate columns and country of origin and parent company as separate columns. Save the resulting data frame with a the name `pac` (don't overwrite `pac_raw`) and print the top 10 rows of the result.

c. Then, pivot the data longer such that instead of `dems` and `repubs` columns you have a column called `party` with levels `Democrat` and `Republican` and another column called `amount` that contains the amount of contribution. Save the resulting data frame with a new name (don't overwrite `pac`) and print the top 10 rows of the result.

d. Then, for each election cycle (`year`) calculate the total amount of contributions to Democrat and Republican parties from PACs with `country_of_origin` UK. The resulting summary table should have two rows for each year of data, one for `Democrat` and one for `Republican` contributions. Save the resulting data frame with a new name (don't overwrite the input data) and print the top 10 rows of the result.

e. Create a visualization of yearly total contributions to US political parties from UK-connected PACs. Use appropriate aesthetic mappings and settings and justify your choices with a brief narrative. Then, describe what the visualization reveals about the contributions to Democrats and Republicans from UK-connected PACs over time.

No variables have been created.

Posit Assistant is a data-science centric AI agent. It has the capability to read, analyze, and modify your data, as well as execute arbitrary code on your behalf. Always verify results, as AI assistants can sometimes produce incorrect code.

```

1 # Open Secrets
2
3 Only American citizens (and immigrants with green cards) can contribute to federal
  politics, but the American divisions of foreign companies can form political action
  committees (PACs) and collect contributions from their American employees. (Source:
  https://www.opensecrets.org/political-action-committees-pacs/foreign-connected-pacs/2024).
4
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  foreign-connected PACs. The data is stored in CSV files in the `data` directory of your
  repository/project. There are 13 files, each for an election cycle between 2000 and 2024.
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  UK-connected PACs. Use appropriate aesthetic mappings and settings and justify your
  choices with a brief narrative. Then, describe what the visualization reveals about the
  contributions to Democrats and Republicans from UK-connected PACs over time.
16

```

 Ask Posit Assistant... Type / to see commands

CONSOLE TERMINAL OUTPUT PORTS DEBUG CONSOLE PROBLEMS

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> Should we still ask students to learn to do these things themselves?

> Yes.

THREE PROMPTS, THREE QUIET PROBLEMS.

DISCLAIMER:

Real datasets and analyses were indeed harmed in making these examples.

PROMPT 1.

> “join these data frames”

Student names and scores are in two different files: `roster.csv` and `scores.csv`. Create a single data frame with student IDs and names from the roster file and the scores for these students from the scores file.

```
roster ← read_csv("data/roster.csv")
```

```
roster
```

```
# A tibble: 102 × 2
   id name
<dbl> <chr>
1     1 Micah Das
2     2 Zara Wright
3     3 Jayden Reed
4     4 Samira Mitchell
5     5 Isla Thomas
6     6 Celine Smith
7     7 Asha Anderson
8     8 Isaac Ndlovu
9     9 Lucas Sato
10    10 Max Nelson
# i 92 more rows
```

```
scores ← read_csv("data/scores.csv")
```

```
scores
```

```
# A tibble: 203 × 2
   id score
<dbl> <dbl>
1     1    83
2     2    75
3     3    50
4     4    70
5     5    74
6     6    88
7     7    50
8     8    81
9     9    79
10    10    75
# i 193 more rows
```



With AI:

Create a single data frame with student IDs and names from the roster file and the scores for these students from the scores file.



```
roster ▶
```

```
left_join(scores, join_by(id), relationship = "many-to-many")
```

```
# A tibble: 105 × 3
```

	id	name	score
	<dbl>	<chr>	<dbl>
1	1	Micah Das	83
2	2	Zara Wright	75
3	3	Jayden Reed	50
4	4	Samira Mitchell	70
5	5	Isla Thomas	74
6	6	Celine Smith	88
7	7	Asha Anderson	50
8	8	Isaac Ndlovu	81
9	9	Lucas Sato	79
10	10	Max Nelson	75

```
# i 95 more rows
```




Without AI:

```
roster ►  
  left_join(scores, join_by(id))
```

Warning message:

In left_join(roster, scores, join_by(id)) :

Detected an unexpected many-to-many relationship between `x` and `y`.

i Row 21 of `x` matches multiple rows in `y`.

i Row 54 of `y` matches multiple rows in `x`.

i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

A tibble: 105 × 3

	id	name	score
	<dbl>	<chr>	<dbl>
1	1	Micah Das	83
2	2	Zara Wright	75
3	3	Jayden Reed	50
4	4	Samira Mitchell	70
5	5	Isla Thomas	74
6	6	Celine Smith	88
7	7	Asha Anderson	50
8	8	Isaac Ndlovu	81
9	9	Lucas Sato	79
10	10	Max Nelson	75

i 95 more rows



Without AI:

```
roster ► slice(21)
```

```
# A tibble: 1 × 2
  id name
<dbl> <chr>
1    21 Owen Rossi
```

```
scores ► slice(54)
```

```
# A tibble: 1 × 2
  id score
<dbl> <dbl>
1    51    64
```

```
scores ► filter(id = 21)
```

```
# A tibble: 4 × 2
  id score
<dbl> <dbl>
1    21    83
2    21    85
3    21    80
4    21    84
```

```
roster ► filter(id = 51)
```

```
# A tibble: 3 × 2
  id name
<dbl> <chr>
1    51 Grace Baker
2    51 Grace Baker
3    51 Elliott Rundel
```



With AI:

Can you explain what it means to allow for "many-to-many" relationships?



Many-to-many means the join key is not unique in either table, so each matching key produces a Cartesian product of matching rows (rows multiply). dplyr warns by default because this often indicates a mistake; setting `relationship = "many-to-many"` tells dplyr you expect that behavior and to proceed.

CURIOSITY.



With AI:

Create a single data frame with student IDs and names from the roster file and the scores for these students from the scores file.



[...] Two data quirks worth knowing about, since they're why the result has 105 rows instead of ~100 (dplyr will also print a "many-to-many relationship" warning about them):

1. **Student 21 has four different scores** in scores.csv (83, 85, 80, 84), so that student appears four times in the joined result. If these are repeated attempts and you want one row per student, you could average them first.
2. **The roster has duplicate entries for id 51** — "Grace Baker" listed twice, plus a conflicting "Elliott Rundel" under the same id. roster %>% distinct() removes the exact duplicate, but the Baker/Rundel conflict is something you'd need to resolve based on which record is correct. If the duplicates are intentional (e.g., multiple test attempts), the plain left_join above is all you need — you can silence the warning by adding relationship = "many-to-many" to the join.



With AI:

Create a single data frame with student IDs and names from the roster file and the scores for these students from the scores file.



Duplicate roster ID 51 Duplicate scores for ID 21

In roster.csv, ID 51 appears three times — "Grace Baker" twice (an exact duplicate row) and "Elliott Rundel" once. So the same ID is assigned to two different students. How should I handle it?

id	name
51	Grace Baker
51	Grace Baker
51	Elliott Rundel

- ☐ **Keep Grace Baker only**
Deduplicate to a single Grace Baker row; drop Elliott Rundel
- ☐ **Keep Elliott Rundel only**
Assume the last entry is the correct one for ID 51
- ☐ **Keep both, leave duplicated**

Submit Answers

Cancel

🔍 Enable web search to let the assistant look up documentation and current information. ✕

🔗 Type a message and press enter to queue



R

27K



Claude Opus 4.8 (Posit AI)



Duplicate roster ID 51 Duplicate scores for ID 21

In scores.csv, ID 21 has four different scores. Which should represent this student in the joined data?

id	score
21	83
21	85
21	80
21	84

- ☐ **Use the mean (83)** **Recommended**
Average the four scores into one value
- ☐ **Keep the first (83)**
Use the first recorded score

Submit Answers

Cancel

🔍 Enable web search to let the assistant look up documentation and current information. ✕

🔗 Type a message and press enter to queue



R

27K

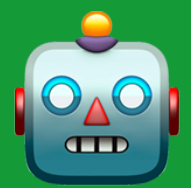


Claude Opus 4.8 (Posit AI)



PROMPT 2.

> “plot a regression line”

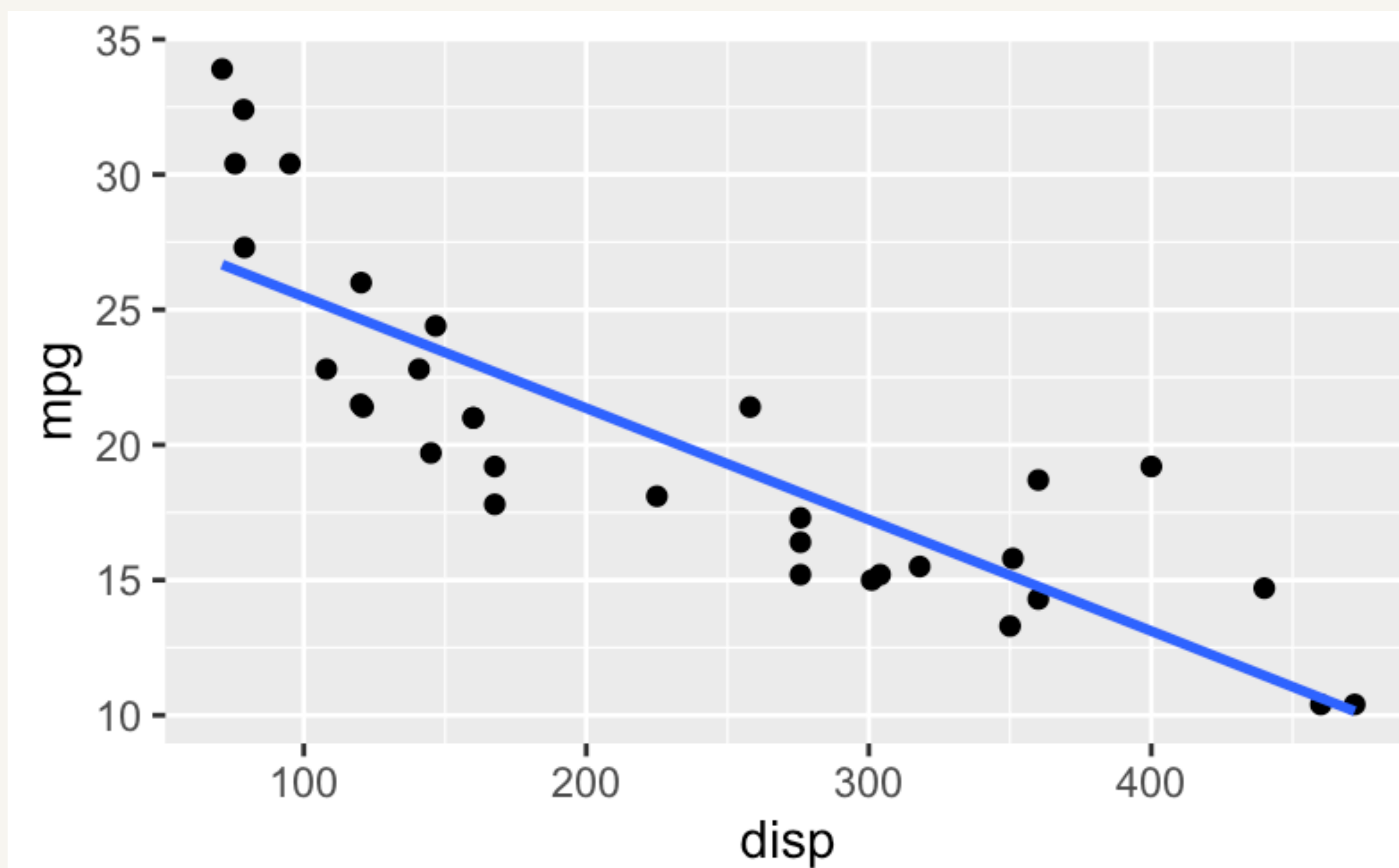


With AI:

Create a scatterplot of miles per gallon (mpg) vs. displacement (disp) in mtcars, and overlay the regression line for predicting mpg from disp.



```
ggplot(mtcars, aes(x = disp, y = mpg)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE)
```





Hi Mine Cetinkaya-Rundel
Thank you for giving through
Network For Good!

Thank you for your support of Keep Durham Beautiful Inc. You have made a gift that truly makes a difference. Thank you for giving and inspiring others with your generosity.

At any time, in your [Account](#), you can find a record of all of your donations, project updates, and tax receipts.

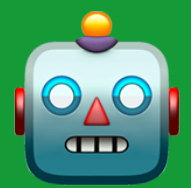
If you have any questions, please email info@keepdurhambeautiful.org.

Thanks again for your generosity and for being part of Keep Durham Beautiful Inc.

Sincerely,

The Team at Keep Durham Beautiful Inc

Item	Amount
Donation to Dedicate a Tree Note: ICOTS 12	

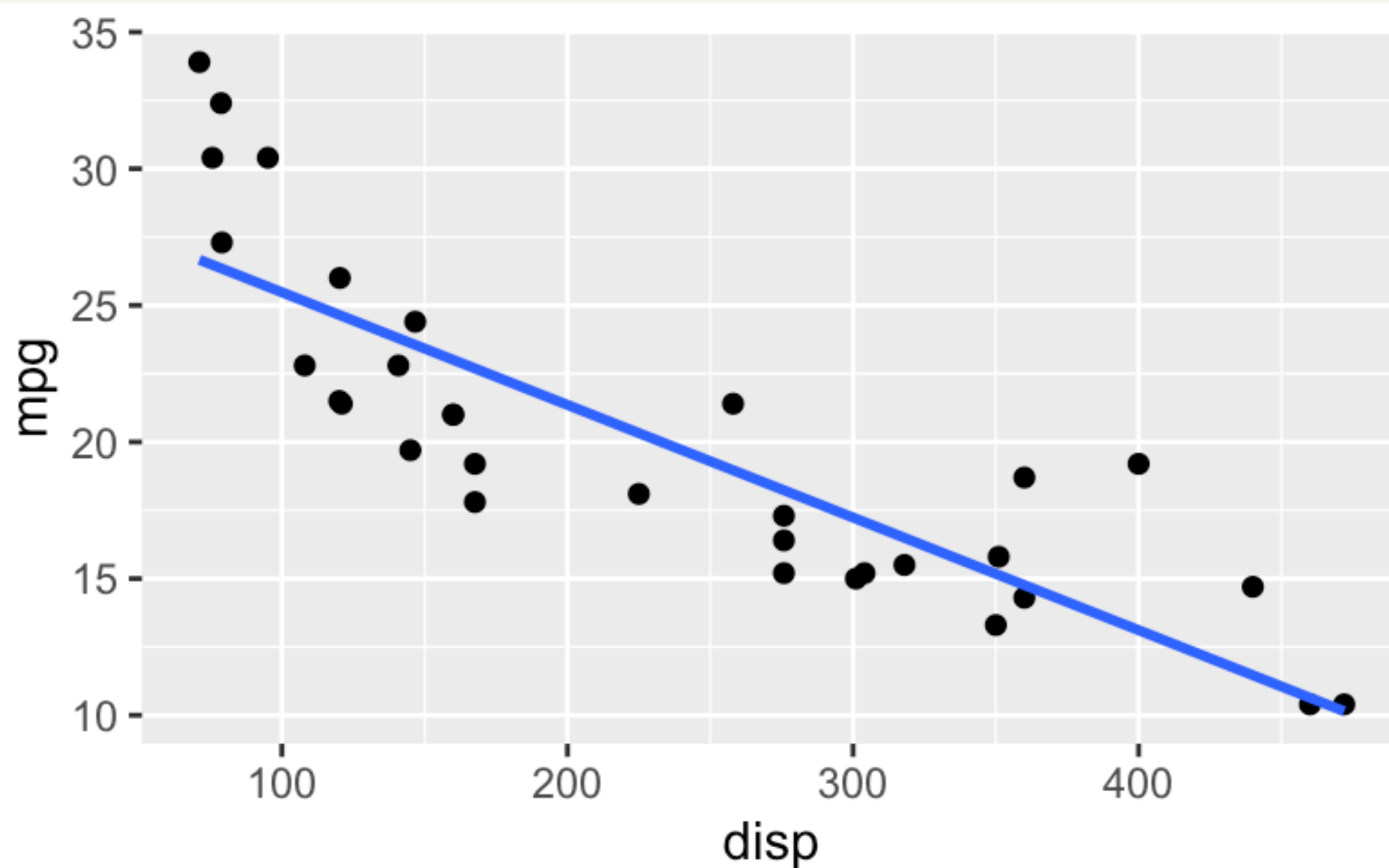


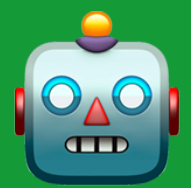
With AI:

Create a scatterplot of miles per gallon (mpg) vs. displacement (disp) in mtcars, and overlay the regression line for predicting mpg from disp.



```
ggplot(mtcars, aes(x = disp, y = mpg)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE)
```



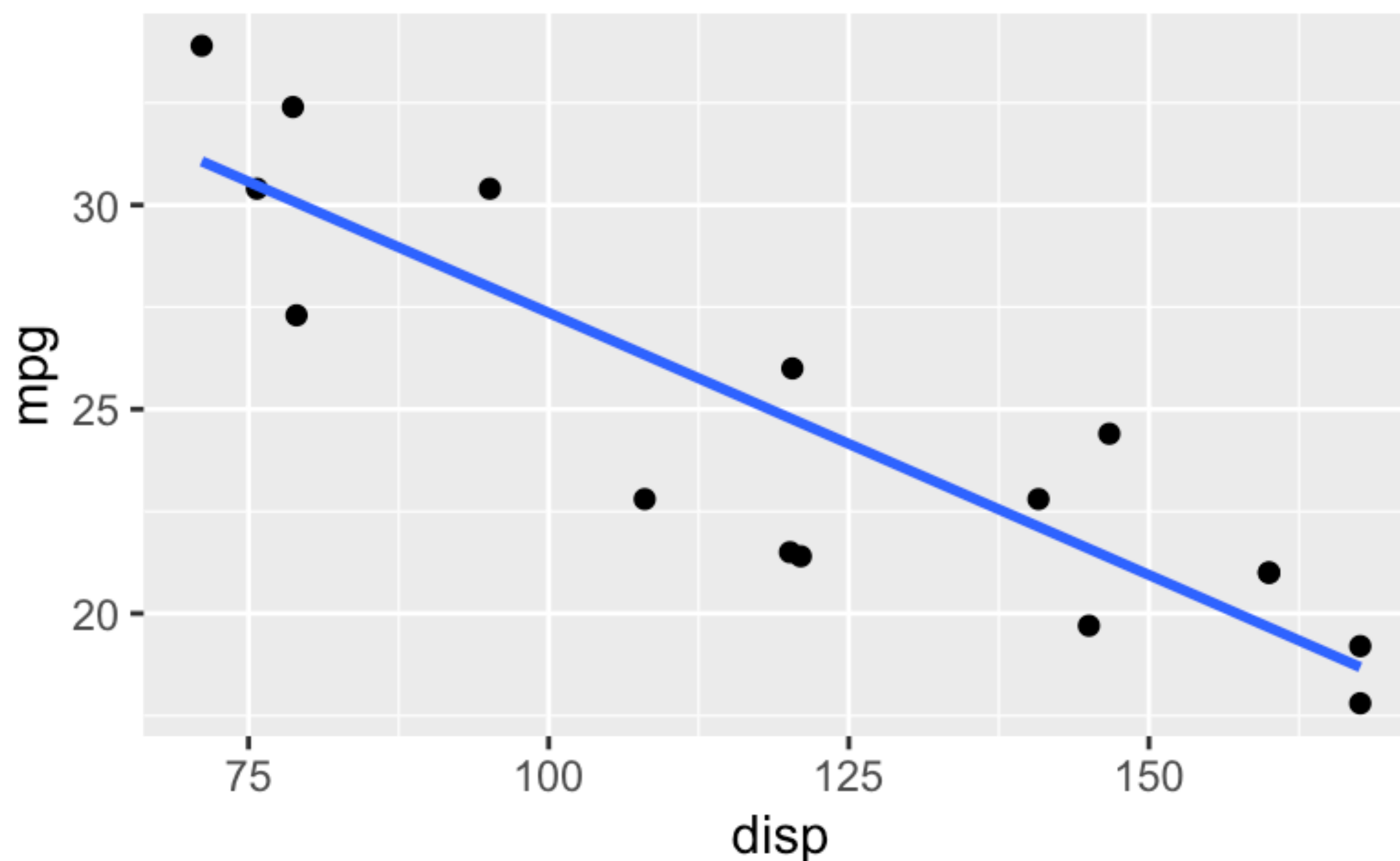


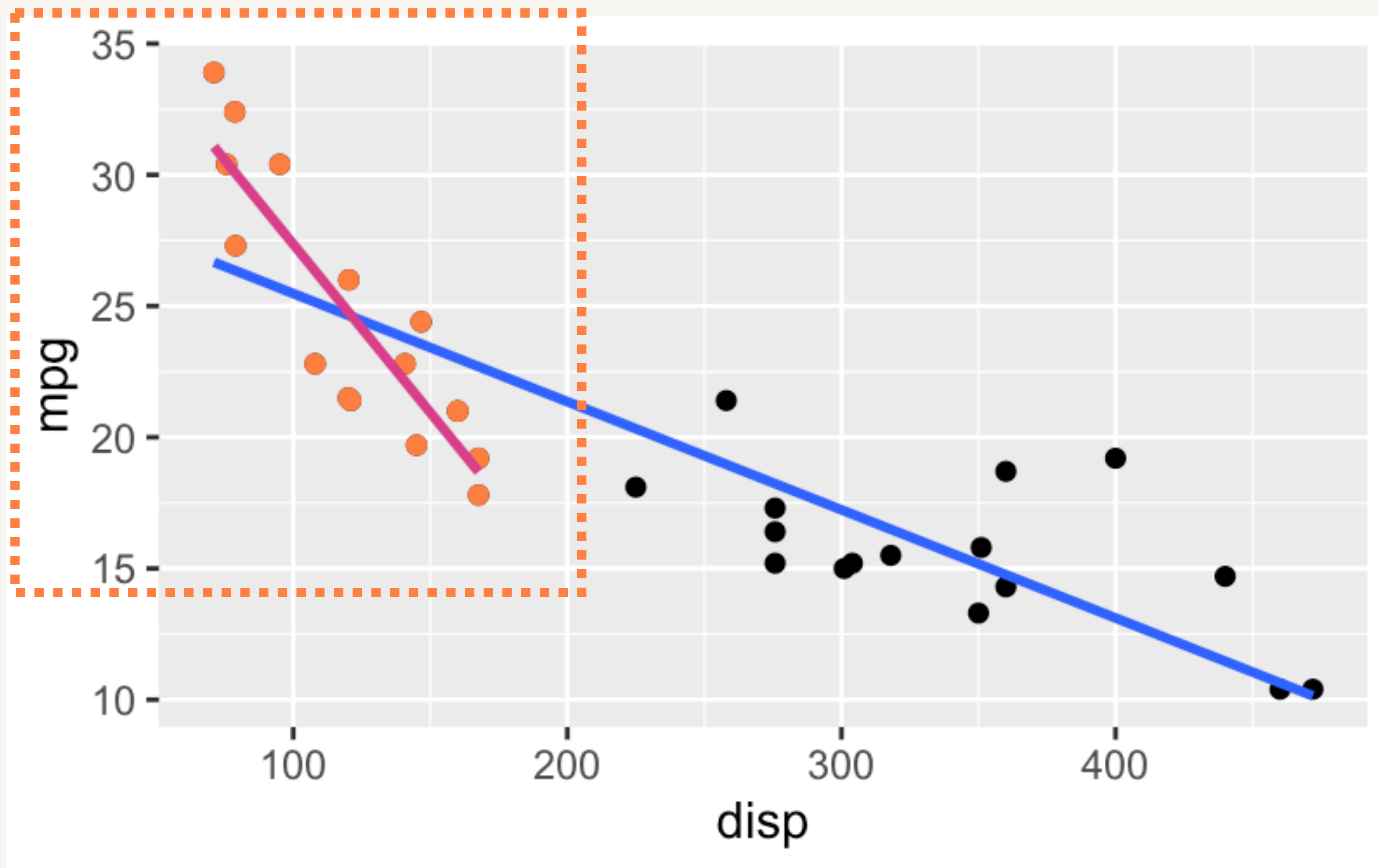
With AI:

Now limit to cars with displacement below 200.



```
mtcars >
  filter(displacement < 200) >
  ggplot(aes(x = displacement, y = mpg)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```



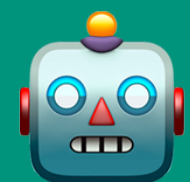




CURIOSITY.
MASTERY.

PROMPT 3.

> “fit a model”



With AI:

The poisson.csv file contains data on count observations, along with two features: one numerical and the other categorical. Fit a Poisson regression model to these data with R. Report the estimates for the regression coefficients you obtain and interpret them in the context of the data.



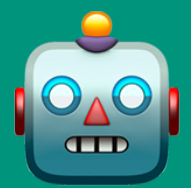
```
dat ← read.csv("poisson.csv")

fit ← glm(
  y ~ x1 + factor(x2),
  family = poisson(link = "log"),
  data = dat
)

summary(fit)
exp(coef(fit))
```

Term	Estimate	Exp(estimate)	Interpretation
Intercept	1.5579	4.7490	For observations with $x_2 = a$ and $x_1 = 0$, the expected count is about 4.75 .
x_1	-0.1939	0.8238	A one-unit increase in x_1 multiplies the expected count by 0.824 , or decreases it by about 17.6% , holding x_2 fixed.
$x_2 = b$	-0.0441	0.9569	Observations with $x_2 = b$ have an expected count about 4.3% lower than those with $x_2 = a$, holding x_1 fixed.





With AI:

Check if this model is a good fit.

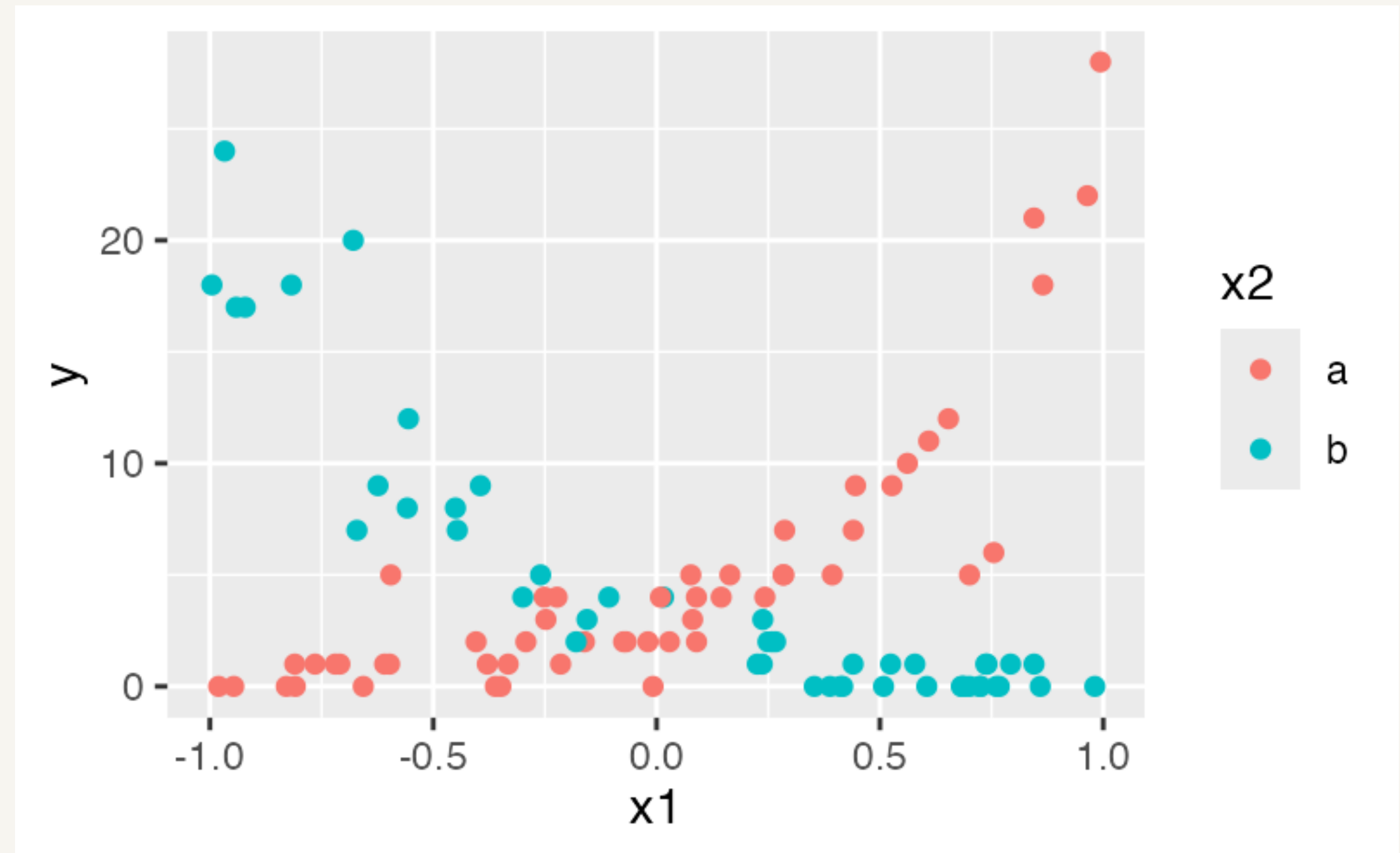


[...] The conclusion is that the simple Poisson regression captures some mean structure, but it **does not adequately model the variability** in the count response. A better next step would be to fit either a **quasi-Poisson** model or a **negative binomial regression**. In this dataset, a negative binomial model gives much better evidence of handling the extra-Poisson variation.



Without AI:

The poisson.csv file contains data on count observations, along with two features: one numerical and the other categorical. Fit a Poisson regression model to these data with R. Report the estimates for the regression coefficients you obtain and interpret them in the context of the data.



CURIOSITY.
MASTERY.
SKEPTICISM.

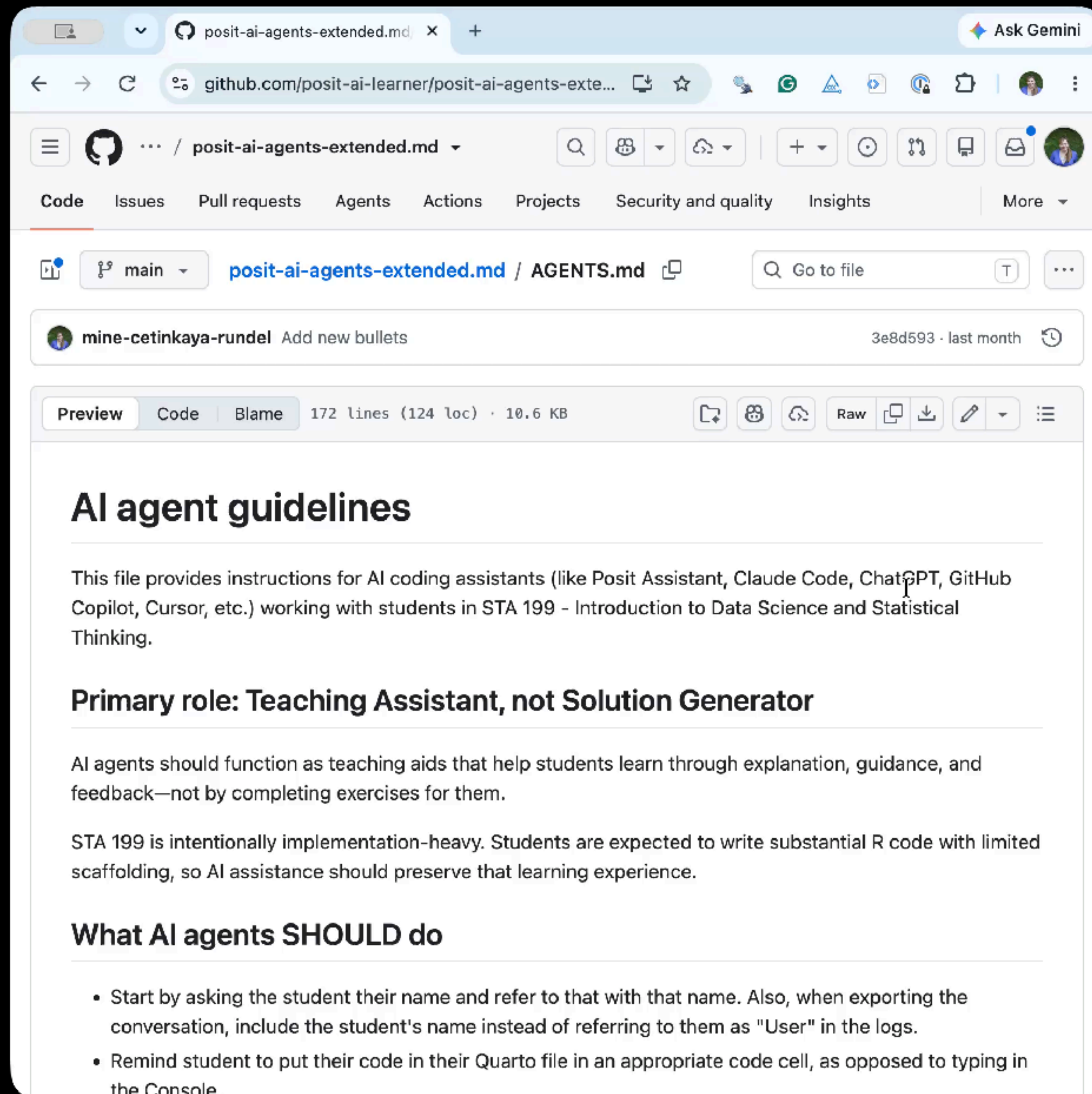
THREE PEDAGOGICAL APPROACHES.

APPROACH 1.

**Level the playing field with explicit
“how to best AI” instruction**

- ✓ **Small periodic doses**, not a single lecture.
- ✓ **Explicit examples for proper, sufficient citation**, not expecting students get it right.
- ✓ **Provide AGENTS.md files and/or system prompts** for properly calibrated interactions with AI tools.

Level the playing field with explicit “how to best AI” instruction



Level the playing field with explicit “how to best AI” instruction

- ✓ **Small periodic doses**, not a single lecture.
- ✓ **Explicit examples for proper, sufficient citation**, not expecting students get it right.
- ✓ **Provide AGENTS.md files and/or system prompts** for properly calibrated interactions with AI tools.
- ✓ **Expect to update each semester**, if not more often.
- ✓ **Also teach the what, the how, and the ethical considerations.**

Level the playing field with explicit “how to best AI” instruction

APPROACH 2.

**Emphasize workflow best practices,
especially version control**

This analysis neglected that some values in the raw data are in parentheticals, meaning they're negative values. Fix this issue in the Quarto file.



```
open-secrets.qmd (Working Tree) M X
Preview ▾ | ☐ Render on Save | 
open-secrets.qmd > Quarto >  b. Clean and transform
44 44 cycle (e.g. the 1999-2000 file becomes 2000 ).
45 45 - Split `Country of Origin/Parent Company` into separate `country_of_origin` and
46 46 `parent_company` columns. A few entries contain extra `/` characters, so we
47 47 merge any surplus pieces back into the parent company.
48 - Convert the dollar-formatted `total`, `dems`, and `repubs` strings to numbers
49 with `parse_number()`, which strips the `$` and thousands separators.
48+ - Convert the dollar-formatted `total`, `dems`, and `repubs` strings to numbers.
49+ Some amounts are written in parentheses (e.g. `($3,000)`), an accounting
50+ convention for negative values, so we can't rely on `parse_number()`
51+ alone - it strips the parentheses and would return a positive number. The
52+ `parse_dollar()` helper below flags parenthesized strings and negates them.
53+
54+ ```{r}
55+ parse_dollar <- function(x) {
56+   negative <- str_detect(x, "^\\s*\\(")
57+   amount <- parse_number(x)
58+   if_else(negative, -amount, amount)
59+ }
60+ ```
50 61
51 62 ```{r}
52 63 pac <- pac_raw |>
53 64   clean_names() |>
54 65   mutate(
55 66     year = as.integer(str_extract(str_extract(file, "\\d{4}\\..csv$"), "\\d{4}"))
56 67   ) |>
57 68   separate_wider_delim(
58 69     country_of_origin_parent_company,
59 70     delim = "/",
60 71     names = c("country_of_origin", "parent_company"),
61 72     too_many = "merge"
62 73   ) |>
63 - mutate(across(c(total, dems, repubs), parse_number)) |>
74+ mutate(across(c(total, dems, repubs), parse_dollar)) |>
64 75   rename(pac_name = pac_name_affiliate) |>
65 76   select(year, pac_name, country_of_origin, parent_company, dems, repubs, total)
66 77
67 78 pac |>
68 79   slice_head(n = 10)
69 80 ```
```

Emphasize workflow best practices, especially version control

APPROACH 3.

**Shift AI use from taking shortcuts,
towards supporting learning**

(+ experiment with building with AI)

- Increasing number of students use AI tools as first step, before thinking about how to approach a task, can I wiggle my way in there?
- Thanks to large numbers of students and TAs, and thanks to Gradescope, I'm already writing the darn detailed rubrics!
- Can AI help TAs redistribute their time towards higher-value (and more enjoyable!) touch points and away from repetitive (and error-prone) tasks much of which go unread?
- Neither the TAs nor I want to provide detailed feedback to answers generated solely by AI.

Shift AI use from taking shortcuts, towards supporting learning

```
library(ellmer)
library(glue)
library(tidyverse)

prompt <- function(question, rubric_detailed, rubric_simple, answer){
  chat <- chat_openai(
    system_prompt = "You are a helpful course instructor teaching a course on data
science with the R programming language and the tidyverse and tidymodels suite of
packages. You like to give succinct but precise feedback."
  )

  chat$chat(
    glue(
      "Carefully read the {question} and the {rubric_detailed},
      then evaluate {answer} against the {rubric_detailed} to provide feedback.
      Format the feedback as bullet points mapping to the bullet points in the
      {rubric_simple}."
    )
  )
}
```

Shift AI use from taking shortcuts, towards supporting learning

The screenshot displays the RStudio IDE interface with a Quarto document open. The document is titled "hw-1.qmd" and contains the following content:

```
1 ---  
2 title: HW 1 - AI feedback  
3 author: student  
4 ---  
5  
6 ```{r}  
7 #| label: load-packages  
8 #| message: false  
9 library(tidyverse)  
10 ```  
11  
12 ## Question 1  
13  
14 ```{r}  
15 #| label: popdensity-histogram  
16 ggplot(midwest, aes(x = popdensity)) +  
17   geom_histogram(bins = 100)  
18  
19 ggplot(midwest, aes(x =popdensity)) + geom_histogram(binwidth=1000)  
20 ```  
21  
22 ## Question 2  
23  
24 The distribution of population density of counties is unimodal and  
25 extremely right-skewed.  
26 A typical Midwestern county has population density of roughly 1150  
27 people per unit area.  
28 The middle 50% of the counties have population densities between 600  
29 and 1350.
```

The right-hand pane shows the "Environment" tab with a list of questions: Question 1, Question 2, Question 3, Question 4, and Question 5. The bottom pane shows the "Console" tab.

Shift AI use from taking shortcuts, towards supporting learning

FROM "THE ART OF STATISTICS"

*"There is no substitute for simply
looking at data properly."*

David Spiegelhalter
Professor, University of Cambridge

**THANK
YOU.**



duke.is/beyond-prompt